

Topic 03: Hypothesis Testing & Decision Making

Statistical Foundations & Probability

TOPIC 03

Hypothesis Testing & Decision Making

The skeptic's null hypothesis, p-values, Type I/II errors, and statistical power.

Cognitive Objectives

By the end of this topic, learners will be able to:

- **Formulate testable hypotheses** with the Null Hypothesis (H_0) as the conservative presumption of zero effect.
- **Interpret p -values accurately** as the probability of observing data at least as extreme under H_0 (not the probability that H_0 is true).
- **Balance Type I (α) vs Type II (β) errors** to maximize statistical power ($1 - \beta$) in production A/B tests.
- **Execute non-parametric permutation tests** in NumPy to evaluate metrics without asymptotic normality assumptions.

The A/B Testing Dilemma

The Business Question: Real or Luck?

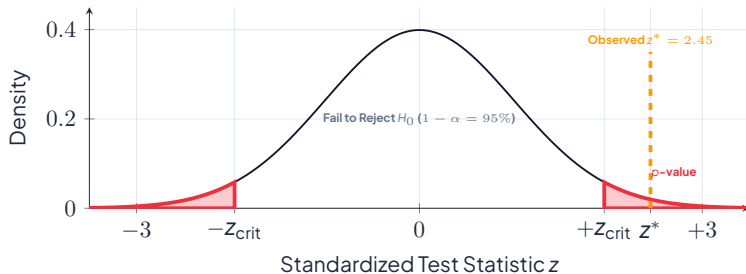
- Control (A): 2,000 users $\rightarrow$ 2.1% conversion.
- Treatment (B): 2,000 users $\rightarrow$ 2.5% conversion (+0.4% lift).
- Product Manager wants to ship immediately!
- **Socratic Probe:** Is this +0.4% lift a real behavioral change or just random coin-flip variance?

The "Aha!" Discovery: Proof by Contradiction

- **Null Hypothesis (H_0):** Treatment has **zero effect** ($\mu_B - \mu_A = 0$). Presumption of innocence!
- **Simulate the Null Universe:** What is the distribution of differences if Variant B is completely useless?
- **p-Value:** How rarely would random noise generate a lift $\geq +0.4\%$?

PUNCHLINE p-value measures surprise under the null hypothesis.

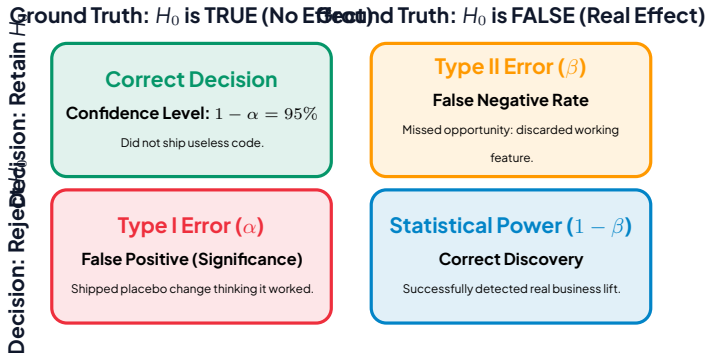
Hero Visualization: Null Distribution & Rejection Region



PUNCHLINE

Reject H_0 if observed z^* falls into the critical $\alpha = 0.05$ extreme tail.

Hero Visualization: Type I vs Type II Decision Matrix



Production Checkpoint: Non-Parametric Permutation Testing

Permutation Test Algorithm

1. Compute observed metric delta $\Delta^* = \bar{B} - \bar{A}$.
2. Pool all observations ($N_A + N_B$).
3. Reshuffle group labels $K = 10,000$ times.
4. Fraction of shuffled trials with $\Delta_{\text{null}} \geq \Delta^*$ gives the exact empirical p -value.

PUNCHLINE Permutation tests eliminate all parametric assumptions.

NumPy Implementation

```
1 import numpy as np
2
3 def permutation_test(ctrl, trt, k=10_000):
4     obs_diff = np.mean(trt) - np.mean(ctrl)
5     pooled = np.concatenate([ctrl, trt])
6     n = len(ctrl)
7     null_diffs = np.empty(k)
8     for i in range(k):
9         np.random.shuffle(pooled)
10        null_diffs[i] = np.mean(pooled[n:]) - np.mean(
11            pooled[:n])
12    return float(np.mean(null_diffs >= obs_diff))
```